

SIMATS ENGINEERING
CSA0310-DATA STRUCTURE
LAB
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1. INTEGER ARRAY

```
Enter number of elements: Enter elements:
```

```
Using Array Indexing:
```

```
arr[0] = 10
```

```
arr[1] = 20
```

```
arr[2] = 30
```

```
arr[3] = 40
```

```
Using Pointer Notation:
```

```
*(ptr+0) = 10
```

2. FACTORIAL

```
Factorial of 6 is 720
```

3. ARRAY ROTATION

```
Enter 10 numbers: Rotate Left (L)/Right (R): By positions: Original: [1996429296  
459 1915950345 32759 1 0 69 0 1996429264 459]
```

```
Step1: Reverse 0-9 -> [459 1996429264 0 69 0 1 32759 1915950345 459 1996429296]
```

```
Step2: Reverse 10-9 -> [459 1996429264 0 69 0 1 32759 1915950345 459 1996429296]
```

```
Step3: Reverse all -> [1996429296 459 1915950345 32759 1 0 69 0 1996429264 459]
```

```
Sum difference: 0
```

4.SPARE MATRIX

```
0 8 0
8 0 0
0 0 9
```

5.FIBONACCI SERIES

```
Enter number of terms: Fibonacci series: 0 1 1 2 3 5 8 13
```

6.ARRAY

```
50
```

7.FIBONACCI SERIES(N>0)

```
Enter number of terms (1-50): Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
Sum of series: 88
Largest term: 34
```

8.UNSIRTED ARRAY

```
Missing elements: 5 6 8
```

9.FIB SERIED ITERATIVE

```
Enter N: Iterative: 0 1 1 2 3 5 8  
Recursive: 0 1 1 2 3 5 8  
Recursive Calls: 59  
Iterative Time: 0.000 ms  
Recursive Time: 0.000 ms
```

10.STRUCUTURE

```
Enter number of students: Enter Roll No Name Marks: Enter Roll No Name Marks:  
Highest Mark: 0 ()  
Class Average: 0.00  
2 records stored successfully.
```